Name_____ Date_____ Period_____

LESSON 6-7 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Factors and Greatest Common Factor

Maryland Standard 6.NS.B.4 · Learning target – I can list the factors of two numbers and find their greatest common factor.

REMEMBER

- A **factor** divides a number with no remainder. Find factors in pairs: 18 = 1 × 18 = 2 × 9 = 3 × 6.
- The **greatest common factor** (GCF) is the greatest factor that two numbers share.
- Use the GCF to make the **greatest number of identical groups** with nothing left over.

WORKED EXAMPLE

Find the GCF of 24 and 36.

Step 1 Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Step 2 Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

Step 3 Common factors: 1, 2, 3, 4, 6, 12. The greatest is 12.

Answer **GCF = 12**

WORD BANK

factor – a number that divides another number evenly

common factor – a factor both numbers have

GCF – the biggest factor both numbers share

prime – a number with only two factors: 1 and itself

PRACTICE

show your work · label units

1. List all the factors of each number.

Step 1 Start with 1 and the number itself.

Step 2 Try 2, 3, 4, … and write each pair that multiplies to the number.

a. 30: _____

b. 28: _____

c. 45: _____

d. 17: _____

e. 48: _____

f. 42: _____

2. Find the GCF of each pair.

Step 1 List the factors of each number.

Step 2 Circle the factors that are in both lists.

Step 3 Pick the biggest circled factor.

a. 12 and 20: ____ **b.** 18 and 27: ____ **c.** 16 and 40: ____

d. 14 and 25: ____ **e.** 36 and 48: ____ **f.** 30 and 45: ____

3. Write the prime factorization of each number. Use exponents.

a. 36 = _____ **b.** 50 = _____ **c.** 72 = _____

4. **Snack Bags.** Mr. Diaz has 36 granola bars and 54 juice boxes. Every bag must be the **same**. Nothing is left over. What is the **greatest** number of bags? What goes in each bag?

Step 1 Find the GCF of 36 and 54. That is the number of bags.

Step 2 Divide each number by the GCF to fill one bag.

Show your work.

Answer_____

5. **Find the Error.** Hana says the GCF of 15 and 28 is 0 because they share no factors. Explain her mistake and give the GCF.

Frame *Hana is wrong because every number has the factor ___. So the GCF of 15 and 28 is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home to name two numbers. Find their GCF together.

Name_____ Date_____ Period_____

LESSON 6-7 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Multiples and Least Common Multiple

Maryland Standard 6.NS.B.4 · Learning target – I can list multiples of two numbers and find their least common multiple.

REMEMBER

- **Multiples** of a number come from skip-counting: 6, 12, 18, 24, …
- The **least common multiple** (LCM) is the least number that is a multiple of both.
- Use the LCM when events **happen together again**, or to buy packs for **equal amounts**.

WORKED EXAMPLE

Find the LCM of 8 and 12.

Step 1 Multiples of 12: 12, 24, 36, …

Step 2 12 is not a multiple of 8. 24 is (8 × 3 = 24).

Answer **LCM = 24**

WORD BANK

multiple – a number you get when you skip-count

common multiple – a multiple both numbers have

LCM – the smallest multiple both numbers share

PRACTICE

show your work · label units

1. Find the LCM of each pair.

Step 1 Skip-count by the bigger number.

Step 2 Stop at the first number the smaller number also divides.

a. 3 and 7: ____

b. 4 and 10: ____

c. 6 and 8: ____

d. 5 and 15: ____

e. 9 and 12: ____

f. 10 and 25: ____

2. **Cookout.** Hot dogs come in packs of 8. Buns come in packs of 10. Malik wants the **same number** of hot dogs and buns. What is the **fewest** packs of each he can buy?

Step 1 Find the LCM of 8 and 10.

Step 2 Divide the LCM by 8 and by 10 to count packs.

Show your work.

Answer_____

3. Blinking Lights. One light blinks every 6 seconds. Another blinks every 9 seconds. They just blinked together. In how many seconds will they blink together again?

Step 1 List multiples of 6 and of 9.

Step 2 Find the first number on both lists.

Answer_____ seconds

4. Two runners start together. One laps the track every 4 minutes, the other every 6 minutes. Which tells when they are next at the start together?

□ GCF of 4 and 6 □ LCM of 4 and 6

5. Reason. Is the LCM of two numbers always their product? Use 4 and 5, and 4 and 6, to explain.

Frame *LCM of 4 and 5 = ___; 4 × 5 = ___. LCM of 4 and 6 = ___; 4 × 6 = ___. So the LCM is ___ the product.*

_____

_____

TEST PRACTICE

choose the best answer

6. What is the LCM of 6 and 15?

A 3 **B** 30 **C** 90 **D** 21

7. Which pair of numbers has an LCM of 24?

A 4 and 6 **B** 6 and 8 **C** 2 and 12 **D** 24 and 48

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___